IDEATHON 2026

PROBLEM STATEMENT TITLE: ANTI-AI:: DEEPFAKE PROTECTOR

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INSTITUTION: AMITY NOIDA

THEME : EMERGING TECHNOLOGIES

PROBLEM STATEMENT

The rapid advancement of generative AI technologies such as Deepfake and Generative Adversarial Network has made it extremely easy to create highly realistic but fake images, videos, and audio.

These deepfakes are being misused for:

- Political misinformation
- Financial fraud & identity theft
- Cyberbullying and defamation
- Fake news propagation
- Biometric spoofing

Existing detection systems:

- Work only for specific media types (image OR video OR audio)
- Fail against new-generation AI models
- Lack real-time adaptability
- Do not provide explainable results
- Cannot ensure authenticity tracking

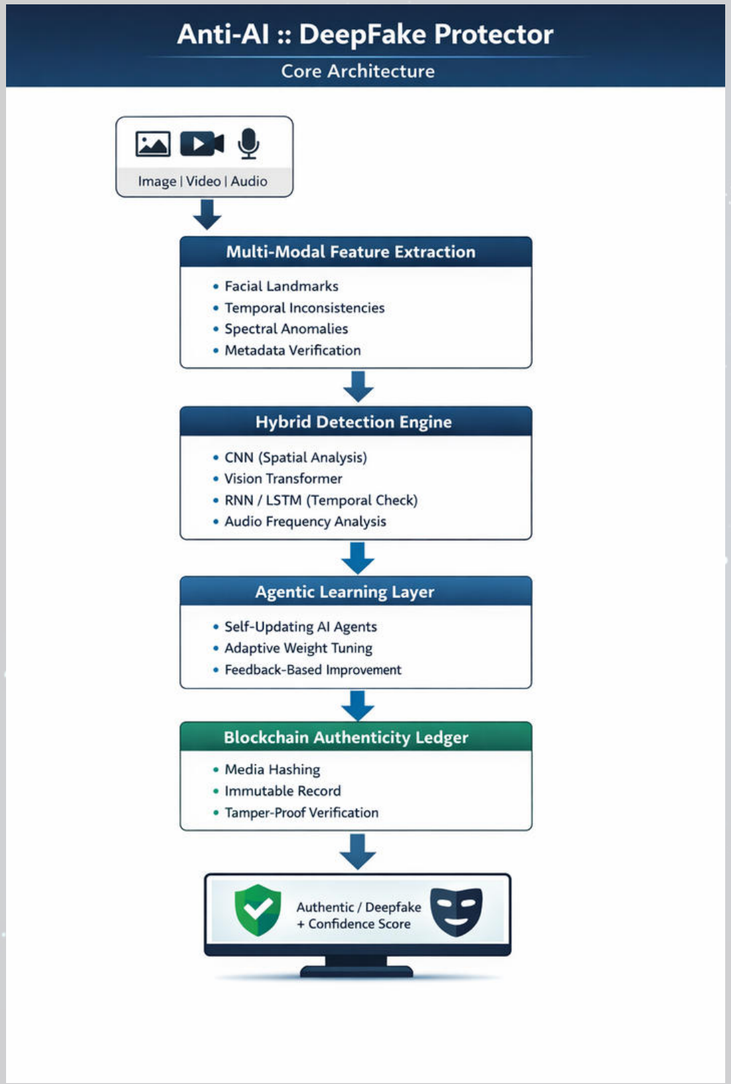
There is a critical need for a multi-modal, adaptive, real-time, and explainable DeepFake detection and protection system that evolves with emerging AI threats.

PROPOSED SOLUTION

An Agentic Multi-Modal Quantum-Inspired AI System that:

- **Detects deepfakes across image, video, and audio**
- **Uses hybrid AI (CNN + Transformer + Spectral Analysis)**
- **Continuously learns from new attack patterns**
- **Provides explainable authenticity scoring**
- **Secures verified media via blockchain-backed AI fingerprinting**

PROPOSED SOLUTION



Core Architecture:

Multi-Modal Feature Extraction

- Facial landmarks
- Temporal inconsistencies
- Spectral anomalies
- Metadata verification

Hybrid Detection Engine

- CNN for spatial inconsistencies
- Vision Transformers for contextual mismatch
- RNN/LSTM for temporal manipulation
- Audio frequency domain analysis

Agentic Learning Layer

- Self-updating detection agents
- Adaptive weight tuning
- Feedback-based model improvement

Blockchain-Based Authenticity Ledger

- Immutable media fingerprint
- Hash verification
- Tamper-proof audit trail

KEY FEATURES / INNOVATION

Parameter	Existing Systems	**Proposed: Anti-AI DeepFake Protector**
Media Support	Single-modal (Image or Video only)	Multi-modal (Image + Video + Audio)
Adaptability	Static trained models	Agentic Self-Learning AI
Detection Technique	CNN-based only	Hybrid CNN + Transformer + Spectral AI
Real-time Processing	Limited	Quantum-Inspired Parallel Processing
Explainability	Black-box output	Confidence Score + Manipulation Heatmaps
Blockchain Integration	Not available	Immutable authenticity verification
Attack Resistance	Weak against latest GANs	Adversarially trained robust model
Scalability	Server dependent	Cloud + Edge deployment ready
Accuracy Improvement	Fixed	Feedback-driven weight tuning
Threat Intelligence	Manual update	Continuous threat adaptation

TARGET BENEFICIARIES

👨‍💼 GOVERNMENT & LAW ENFORCEMENT

- FAKE POLITICAL PROPAGANDA DETECTION
- CYBERCRIME INVESTIGATION

📰 MEDIA & JOURNALISM HOUSES

- VERIFICATION OF VIRAL CONTENT
- FAKE NEWS FILTERING

🏦 BANKING & FINANCIAL INSTITUTIONS

- KYC VERIFICATION
- BIOMETRIC FRAUD PREVENTION

🏢 CORPORATE ENTERPRISES

- BRAND PROTECTION
- EXECUTIVE IMPERSONATION PREVENTION

🎓 EDUCATIONAL INSTITUTIONS

- ACADEMIC INTEGRITY MONITORING

👤 GENERAL PUBLIC

- SOCIAL MEDIA AUTHENTICITY
- VERIFICATION

DELIVERABLES / PROPOSED OUTCOME

🔹 **Technical Deliverables**

- Fully functional web-based DeepFake detection system
- Real-time upload & detection interface
- Multi-modal AI detection engine
- Blockchain-based authenticity verification module
- Dashboard with analytics & reporting
- API for third-party integration

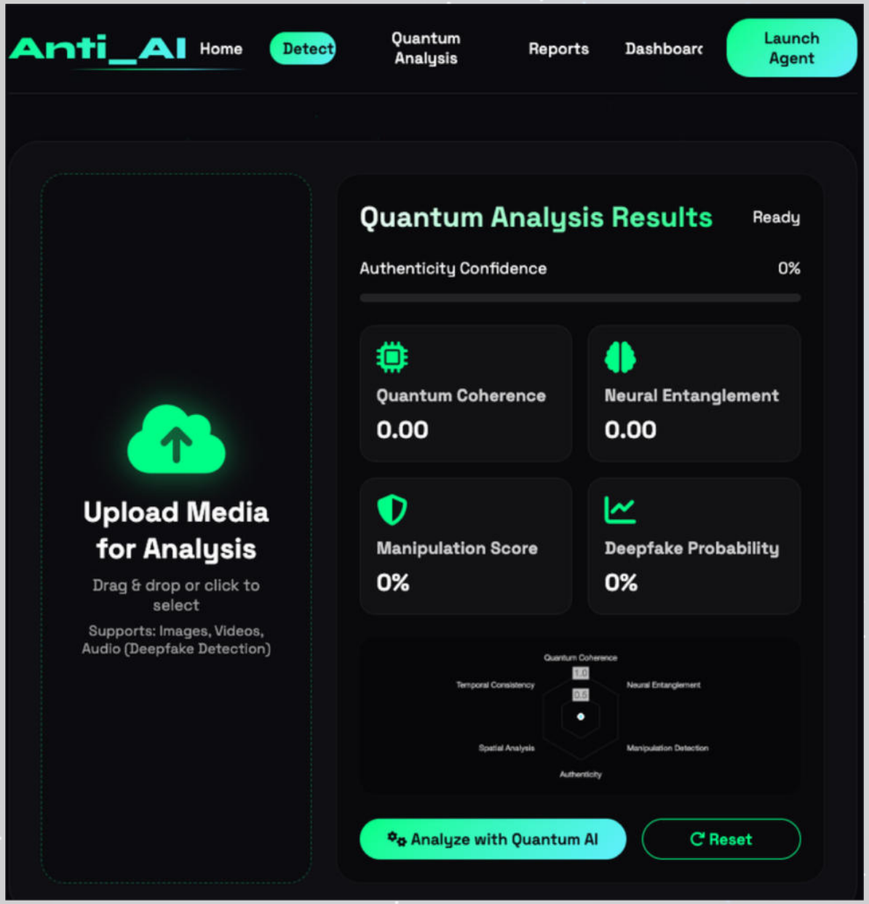
🔹 **Measurable Outcomes**

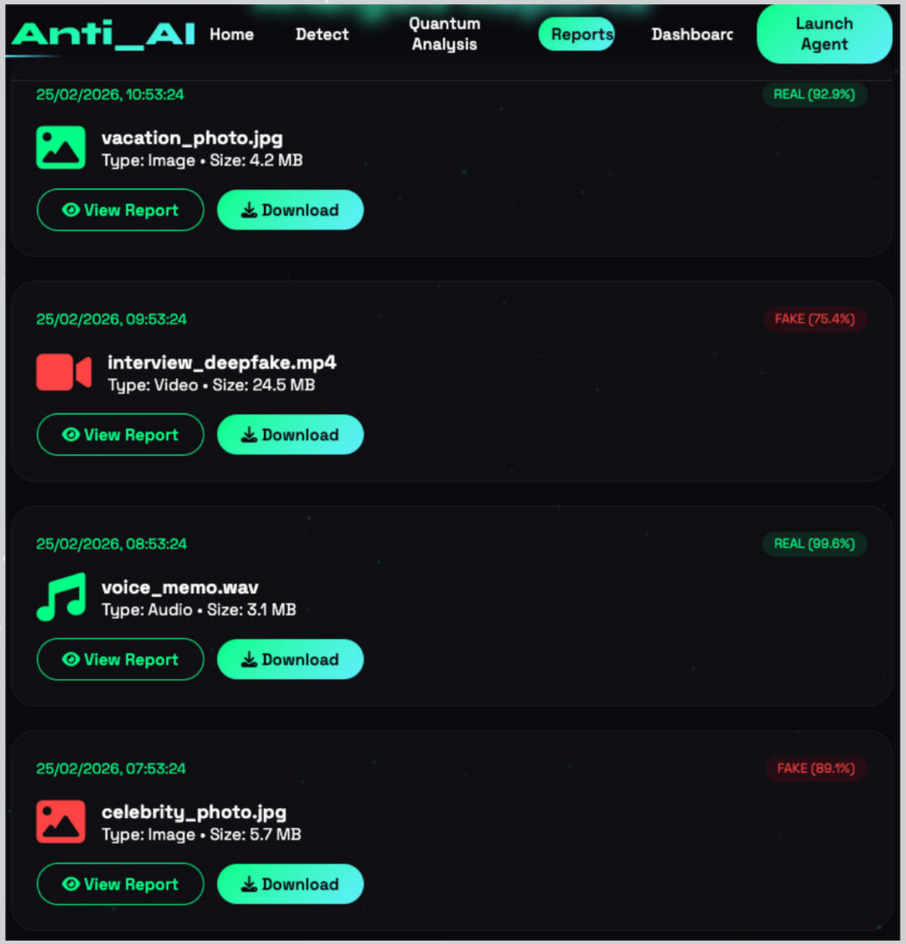
- ≥ 97% detection accuracy (target benchmark)
- Real-time detection (< 5 seconds per file)
- Reduced false positives
- Scalable architecture for large datasets
- Continuous model improvement through feedback loop

🔹 **Social Impact**

- Reduced digital misinformation
- Increased trust in digital media
- Enhanced cybersecurity resilience
- Protection against identity fraud

PROTOTYPE
(IMG OR VID IF ANY)





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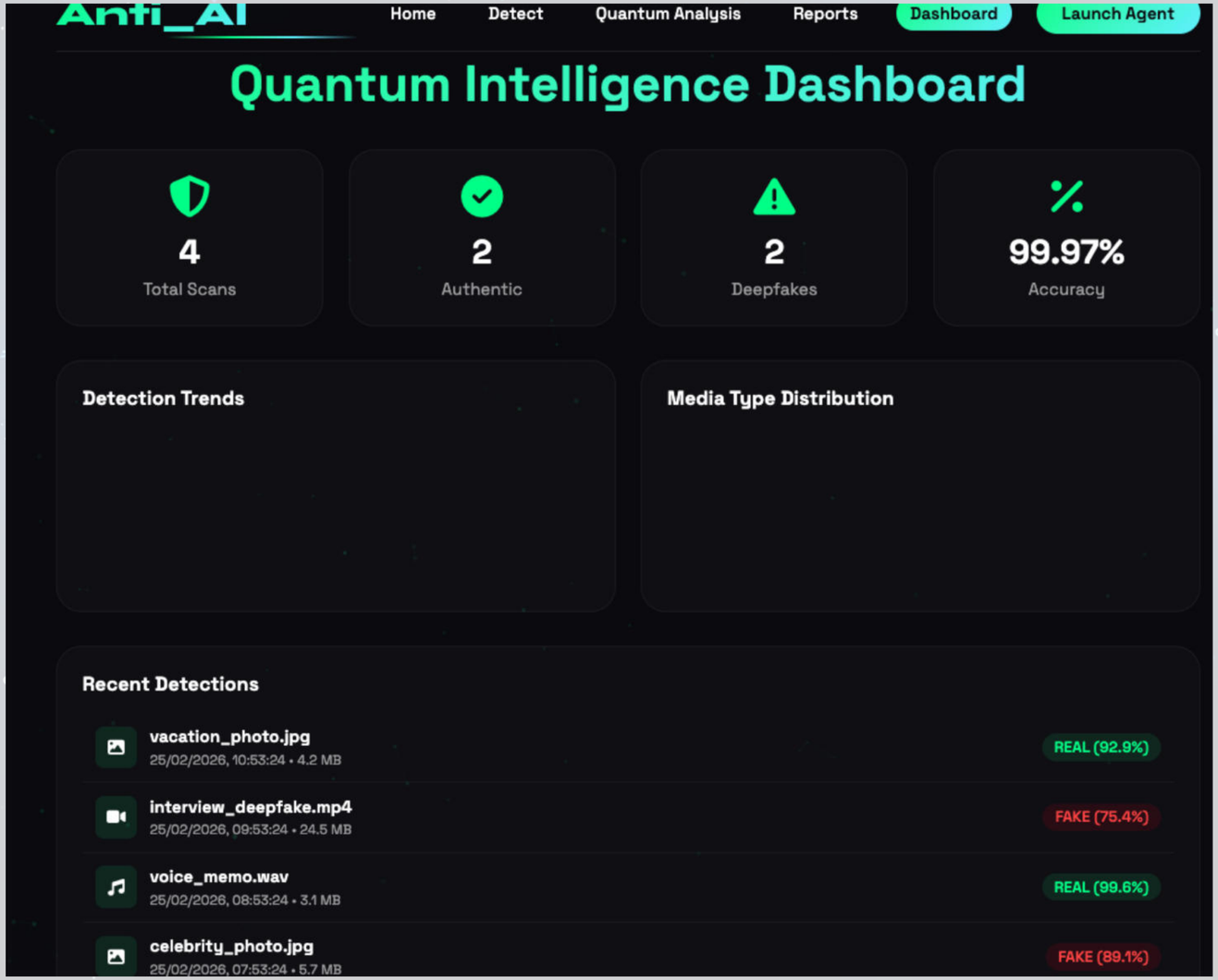

THANK YOU